# Joseph Li

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## SKILLS

**Languages:** C++, C, Python, Go, SQL, CUDA

**Systems & Tools:** Linux, QEMU, Git, GitHub Actions, CI/CD, Docker, Kubernetes, PostgreSQL, Redis

**Concepts:** Operating Systems, Virtual Memory, File Systems, Buffer Management, Process Scheduling, Benchmarking, Regression Testing, Quantization, PyTorch

## PROFESSIONAL EXPERIENCE

**CORTEX AI** — **Menlo Park, CA**

***AI Engineer Intern*** — ***Jun. 2026 - Aug. 2026***

- Designed and launched an LLM API gateway in **Go**, **PostgreSQL**, and **Redis** that routes OpenAI, Anthropic, and **AWS Bedrock** through one endpoint for a 2,000-seat customer and meters $10K in monthly model spend
- Built a pre-release test suite with **GitHub Actions**, expanding coverage from 5 smoke tests to 159 tasks so tool regressions were caught before release
- Established live monitoring in **Langfuse**, adopted by all 9 engineers, tracking per-session latency and failures and routing each failure to the engineer responsible
- Reduced wasted model spend by stopping runaway agent loops and using an **LLM judge** to flag unnecessary tool calls

**YOYO MARKETS INC.** — **New York, NY (Remote)**

***AI Engineer*** — ***Aug. 2025 - Dec. 2025***

- Automated manual research and editorial work with a **Python** and LLM generation pipeline, producing **50+ markets per week at 5x the hand-curated rate**
- Developed an **asynchronous Python** scraper tracking 150+ high-signal Twitter accounts in real time, feeding that pipeline **3x the signal volume** of public trend aggregators
- Devised a ranking step that uses pairwise LLM comparisons and Elo ratings to screen 200+ candidate markets per cycle and send the strongest to founder review
- Shipped a **Next.js** review dashboard on **Vercel** that founders use daily, cutting **publishing turnaround from 30 minutes to under 5 minutes**

## RESEARCH AND PROJECTS

**EFFICIENT LLMS** | *PyTorch, CUDA, LoRA* — **Jan. 2026 - May 2026 | USC**

- Converted **Llama 3.2-1B** into a **Mixture-of-Experts** model with **LoRA** adapters and kept held-out quality level with the original, perplexity **9.60 versus 9.66**
- Quantized weights from **FP16** to **INT4**, cutting model size **61%** while keeping generated answers coherent

**WEENIX (A UNIX-LIKE OS KERNEL)** | *C, x86, QEMU* — **Aug. 2025 - Dec. 2025 | USC**

- Constructed a Unix-like kernel in **C**, with process scheduling, a file system, and virtual memory
- Integrated the three parts in **QEMU** so processes could be scheduled, files created and removed, and a copied process shared memory until one side wrote to it

**POSTGRESQL BUFFER MANAGER** | *C, PostgreSQL* — **Jan. 2025 - May 2025 | USC**

- Replaced **PostgreSQL**'s default memory rule with a first-in, first-out rule that discards the oldest unused data
- Benchmarked its performance against PostgreSQL's default implementation and found the simpler rule **matched its speed** at 1MB of memory

## EDUCATION

**UNIVERSITY OF SOUTHERN CALIFORNIA** — **Los Angeles, CA**

*M.S. Computer Science | GPA: 3.6/4.0* — *Sep. 2024 - May 2026*

Relevant Coursework: Operating Systems, ML Accelerators

**UNIVERSITY OF MICHIGAN, ANN ARBOR** — **Ann Arbor, MI**

*B.S.E. Data Science | GPA: 3.7/4.0* — *Sep. 2020 - May 2024*